The documentation and process conversion measures necessary to comply with this revision shall be completed by 3 Nov 1993

INCH-POUND

MIL-S-19500/441D
3 May 1993
SUPERSEDING
MIL-S-19500/441C
6 April 1988

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, POWER
TYPES 2N3740 AND 2N3741,
JANTX, JANTXV, JANS, AND JANC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, silicon, power transistors. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1, TO-213AA (formerly TO-66) and figure 2 (JANC).

1.3 Maximum ratings.

Types	P_T 1/		V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_{STG} and T_J
	T_C = +25°C	T_C = +100°C						
	W	W	V dc	V dc	V dc	A dc	A dc	°C
	25	14			7	2	4	-65 to +200
2N3740			60	60				
2N3741			80	80				

1/ Derate at 143 mW/°C above T_C = +25°C.

1.4 Primary electrical characteristics at T_C = +25°C.

Limits	h_{FE2} 1/	h_{FE4} 1/	$\|h_{fe}\|$	$V_{CE(sat)2}$ 1/	C_{obo}	Pulse response		$R_{\theta JC}$
	V_{CE} = 1 V dc I_C = 250 mA dc	V_{CE} = 1 V dc I_C = 1 A dc	V_{CE} = 10 V dc I_C = 100 mA dc f = 5 MHz	I_C = 1.0 A dc I_B = 125 mA dc	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	t_{on}	t_{off}	
				V dc	pF	ns	µs	°C/W
Min	30	10	1					
Max	120		12	0.6	100	400	1	7

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPO), NASA Goddard Space Flight Center, Code 310-A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The designs, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1 and 2 herein. Current density of internal conductors shall be as specified in MIL-S-19500.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

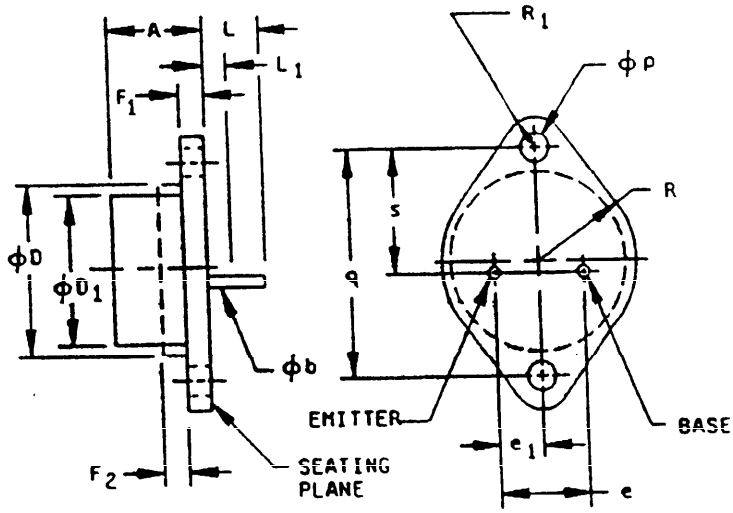
4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 JANC devices. JANC devices are qualified in accordance with appendix H of MIL-S-19500.

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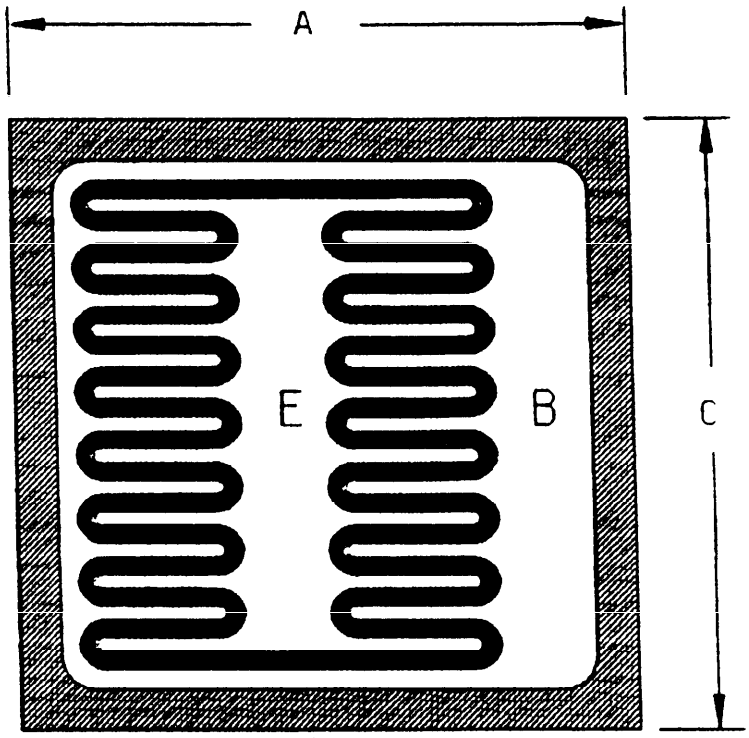
Symbol	Dimensions		Notes
	Inches		
	Min	Max	
A	.250 (6.35)	.340 (8.54)	
ϕb	.028 (0.71)	.034 (0.86)	5, 9
ϕD	—	.620 (15.75)	3
ϕD_1	.470 (11.94)	.500 (12.70)	3
e	.190 (4.83)	.210 (5.33)	4
e_1	.093 (2.36)	.107 (2.72)	4
F_1	.050 (1.27)	.075 (1.91)	3
F_2	—	.050 (1.27)	3
L	.360 (9.14)	.500 (12.70)	5
L_1	—	.050 (1.27)	5, 9
ϕp	.142 (3.61)	.152 (3.86)	8
q	.958 (24.33)	.962 (24.43)	
R	—	.350 (8.89)	
R_1	.115 (2.92)	.145 (3.68)	6
s	.570 (14.48)	.590 (14.99)	4



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Body contour is optional within zone defined by ϕ_D and ϕ_{D1}.
4. These dimensions should be measured at points .050 to .055 below seating plane. When gauge is not used, measurement will be made at seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. The collector shall be electrically connected to the case.
9. ϕ_b applies between L_1 and L. Diameter is uncontrolled in L_1.

FIGURE 1. Physical dimensions, TO-213AA.

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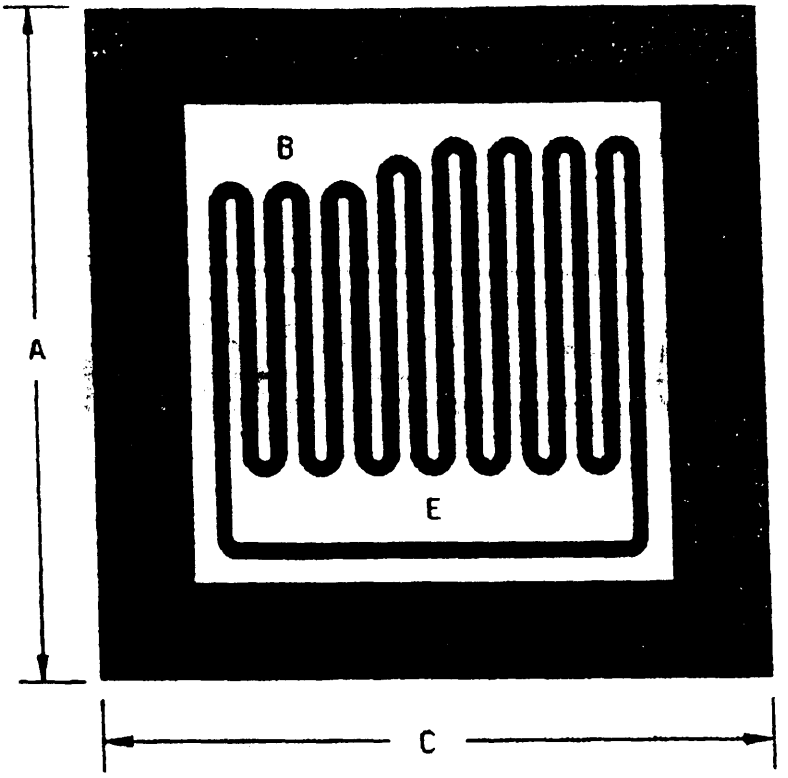
A - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.059	.069	1.49	1.76
C	.059	.069	1.49	1.76

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Thickness: .008 inch (0.20 mm) to .012 inch (0.30 mm). Top metal: Aluminum 25,000 Å minimum, 30,000 Å nominal. Back metal: Gold 2,500 Å minimum, 3,000 Å nominal. Back side: Collector. Bonding pad:
   B = .045 inch (1.14 mm) x .0075 inch (0.19 mm).
   E = .039 inch (0.99 mm) x .0075 inch (0.19 mm).
4. Element evaluation shall be performed on case outline TO-213AA.

FIGURE 2. Physical dimensions, JANC die.

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B - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.083	.087	2.11	2.21
C	.083	.087	2.11	2.21

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Thickness: .014 inch (0.36 mm) to .018 inch (0.46 mm). Top metal: Aluminum 20,000 Å minimum, 30,000 Å nominal 50,000 Å maximum. Back metal: T1 = 5,000 Å nominal ±2,000 Å; N1 = 10,000 Å nominal ±2,000 Å; AG = 10,000 Å nominal ±2,000 Å. Back side: Collector. Bonding pad:
   B = .021 inch (0.53 mm) x .008 inch (0.20 mm) ±.002 inch (0.05 mm).
   E = .040 inch (1.01 mm) x .008 inch (0.20 mm) ±.002 inch (0.05 mm).
4. Element evaluation shall be performed on case outline TO-213AA.

FIGURE 2. Physical dimensions, JANC die - Continued.

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4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-STD-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal response (see 4.3.3)	Thermal response (see 4.3.3)
9	I_{CEX1} and h_{FE2}	Not applicable
11	I_{CEX1} and h_{FE2}; ΔI_{CEX1} = 100 percent of initial value or 50 nA dc, whichever is greater; Δh_{FE2} = ±25 percent of initial value.	I_{CEX1} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX1} = 100 percent of initial value or 50 nA dc, whichever is greater; Δh_{FE2} = ±25 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CEX1} = 100 percent of initial value or 50 nA dc, whichever is greater; Δh_{FE2} = ±25 percent of initial value.

1/ This test shall be performed anytime before screen 9.

4.3.1 Power burn-in conditions. Power burn-in conditions (all levels) are as follows:

T_J = +187.5°C ±12.5°C; V_{CB} = 25 ±5 V dc; $T_A \leq$ +100°C.

4.3.2 Screening (JANC). Screening of JANC die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100 percent probed in accordance with group A, subgroup 2.

4.3.3 Thermal response (ΔV_{BE} measurements). The ΔV_{BE} measurements shall be performed in accordance with MIL-STD-750, method 3131. The ΔV_{BE} conditions (I_H and V_H) and maximum limit shall be derived by each vendor. The chosen ΔV_{BE} measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen ΔV_{BE} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. One-hundred percent safe operating area (SOA) testing may be performed in lieu of thermal response testing herein provided that the appropriate conditions of temperature, time, current, and voltage to achieve die attach integrity are approved by the qualifying activity. The following parameter measurements shall apply:

a. I_H measurement . . . . . 10 mA.

b. V_{CE} measurement voltage . . . . . 16 V (same as V_H).

c. I_H collector heating current . . . . . 1 A (minimum).

d. V_H collector-emitter heating voltage . . . . . 16 V (minimum).

e. t_H heating time . . . . . 10 ms.

f. t_{MD} measurement delay time . . . . . 50 µs.

g. t_{SW} sample window time . . . . . 10 µs (maximum).

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table III herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, (table IVa (JANS) of MIL-S-19500).

a. Subgroup 4: Intermittent operation life (LTPD); method 1037; P_T = 3 W at T_A = +25°C, $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling on the device shall be permitted.

b. Subgroup 5: Steady-state operation life (LTPD); method 1027; $T_A \leq$ +150°C for 96 hours; P_T = 3 W at T_A = +100°C or adjusted as required by the chosen T_A to give an average lot T_J = +275°C.

c. Subgroup 6: Thermal resistance; method 3131; see 4.5.2 herein.

4.4.2.2 Group B inspection, (table IVb (JANTX and JANTXV) of MIL-S-19500). Subgroup 3: Steady-state operation life (LTPD); method 1027; T_J = +187.5°C ±12.5°C, $V_{CE} \geq$ 10 V dc, $T_A \leq$ +100°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C, inspection (table V of MIL-S-19500).

a. Subgroup 2: Terminal strength; method 2036; test condition A, weight = 10 pounds, time = 15 s.

b. Subgroup 6: Steady-state operation life (LTPD); method 1027; $V_{CE} \geq$ 10 V dc, P_T = 14 W at T_C = +100°C or +100°C ≤ T_C ≤ +125°C with P_T varied according to the chosen T_C to achieve a T_J = +187.5°C ±12.5°C.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500 and table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a.	I_M measurement	10 mA.
b.	V_{CE} measurement voltage	16 V dc.
c.	I_H collector heating current	0.5 A, minimum.
d.	V_H collector-emitter heating voltage	16 V dc.
e.	t_H heating time	Steady-state (see MIL-STD-750, method 3131 for definitions) or 1.0s minimum.
f.	t_{MD} measurement delay time	50 to 80 µs.
g.	t_{SW} sample window time	10 µs maximum

4.5.3 Inspection conditions. Unless otherwise specified in MIL-S-19500 or herein, all inspections shall be conducted at a case temperature of +25°C ±3°C.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 100 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3740				60		V dc
2N3741				80		V dc
Collector to emitter cutoff current	3041	Bias condition D	I_{CEO}		10	µA dc
2N3740		V_{CE} = 40 V dc				
2N3741		V_{CE} = 60 V dc				
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 7 V dc	I_{EBO}		100	nA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc	I_{CEX1}		300	nA dc
2N3740		V_{CE} = 60 V dc				
2N3741		V_{CE} = 80 V dc				
Collector to base cutoff current	3036	Bias condition D	I_{CBO}		100	nA dc
2N3740		V_{CB} = 60 V dc				
2N3741		V_{CB} = 80 V dc				
Base emitter voltage (nonsaturated)	3066	Test condition B; V_{CE} = 1 V dc; I_C = 250 mA dc pulsed (see 4.5.1)	V_{BE}		1	V dc
Saturation voltage and resistance	3071	I_C = 250 mA dc; I_B = 25 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.4	V dc
Saturation voltage and resistance	3071	I_C = 1 A dc; I_B = 125 mA dc pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.6	V dc
Forward-current transfer ratio	3076	V_{CE} = 1 V dc pulsed (see 4.5.1)				
		I_C = 100 mA dc	h_{FE1}	40		
		I_C = 250 mA dc	h_{FE2}	30	120	
		I_C = 500 mA dc	h_{FE3}	20		
		I_C = 1 A dc	h_{FE4}	10		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 4 A dc pulsed (see 4.5.1)	h_{FE5}	3		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc	I_{CEX2}		25	µA dc
2N3740		V_{CE} = 60 V dc				
2N3741		V_{CE} = 80 V dc				
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 250 mA dc pulsed (see 4.5.1)	h_{FE6}	10		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 1 kHz	h_{fe}	25	250	
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 100 mA dc; f = 5 MHz	$\lvert h_{fe} \rvert$	1	12	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 KHz ≤ f ≤ 1 MHz	C_{obo}		100	pF
Pulse response						
Turn-on time		I_C = 1 A dc; I_{B1} = 0.1 A dc (see figure 3)	t_{on}		400	ns
Turn-off time		I_C = 1 A dc; $I_{B1} = I_{B2}$ = 0.1 A dc (see figure 3)	t_{off}		1	µs
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; power application time = 1 s, 1 cycle (see figure 4)				
Test 1		V_{CE} = 6.25 V dc; I_C = 4 A dc				
Test 2		V_{CE} = 20 V dc; I_C = 1.25 A dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Test 3						
2N3740		V_{CE} = 50 V dc; I_C = 150 mA dc				
2N3741		V_{CE} = 65 V dc; I_C = 150 mA dc				
Safe operating area (clamped switching)		T_A = +25°C; V_{CC} = 15 V dc; duty cycle ≤ 5 percent, t_p = 1.5 ms (vary to obtain I_C), I_C = 4 A dc (see figures 5 and 6)				
2N3740		Clamp voltage = 60 V dc				
2N3741		Clamp voltage = 80 V dc				
End-point electrical measurements		See table III, steps 1 and 3				
Safe operating area (switching)	3053	Load condition C (unclamped inductive) (see figure 7); T_A = +25°C; duty cycle ≤ 5 percent, R_S = 0.1Ω; $t_r = t_f$ ≤ 500 ns; R_{BB1} = 50Ω; V_{BB1} = 10 V dc; R_{BB2} = ∞; V_{BB2} = 0; V_{CC} ≥ 20 V dc;				
Test 1		t_p = 375 µs (vary to obtain I_C); I_C = 1 A dc; L = 5 mH (min) at 1 A with maximum dc resistance of 0.5Ω. For reference only: 2 ESSEX Stancor C-2688 (in parallel), or equivalent.				
Test 2		t_p = 1.5 ms (vary to obtain I_C); I_C = .25 A dc; L = 80 mH (min) at .25 A with a maximum dc resistance of 1 ohm. For reference only: ESSEX Stancor C-2691 or Triad C = 48µ				
Safe operating area (clamped switching) (destructive)		T_A = +25°C; V_{CC} = 55 V dc (see figures 5 and 6); duty cycle ≤ 5 percent, t_p = 1.5 ms (vary to obtain I_C); I_C = 4 A dc				
2N3740		Clamp voltage = 60 V dc				
2N3741		Clamp voltage = 80 V dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
End-point electrical measurements		See table III, steps 1 and 3				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification inspection
	Method	Conditions	
Subgroup 1			22 devices, c = 0
Temperature cycling	1051	500 cycles	
Electrical measurements		See table III, steps 1, 3, and 6	
Subgroup 2			45 devices, c = 0
Steady-state dc blocking life	1039 or 1049	Conditions A, 500 hours	
Electrical measurements		See table III, steps 1, 4, 5, and 6	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices, c = 0
Thermal resistance	3131	$R_{\theta JC}$ = 7°C/W maximum, see 4.5.2	

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TABLE III. Groups A, B, C, and E electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current 2N3740 2N3741	3041	Bias condition A; V_{BE} = 1.5 V dc V_{CE} = 60 V dc V_{CE} = 80 V dc	I_{CEX1}		300	nA dc
2.	Collector to emitter cutoff current 2N3740 2N3741	3041	Bias condition A; V_{BE} = +1.5 V dc V_{CE} = 60 V dc V_{CE} = 80 V dc	I_{CEX2}		500	nA dc
3.	Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 250 mA dc; pulsed (see 4.5.1)	h_{FE2}	30	120	
4.	Saturation voltage and resistance	3071	I_C = 250 mA dc; I_B = 25 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.4	V dc
5.	Collector to emitter cutoff current 2N3740 2N3741	3041	Bias condition A; V_{BE} = 1.5 V dc V_{CE} = 60 V dc V_{CE} = 80 V dc	ΔI_{CEX1}		100 percent of initial value or 50 nA dc, whichever is greater	
6.	Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 250 mA dc; pulsed (see 4.5.1)	Δh_{FE2}		±25 percent change from initial reading	
7.	Saturation voltage and resistance	3071	I_C = 250 mA dc; I_B = 25 mA dc; pulsed (see 4.5.1)	$\Delta V_{CE(sat)1}$		50 mV dc change from initial value	
8.	Thermal response 4/	3131	See 4.3.3	ΔV_{BE}			

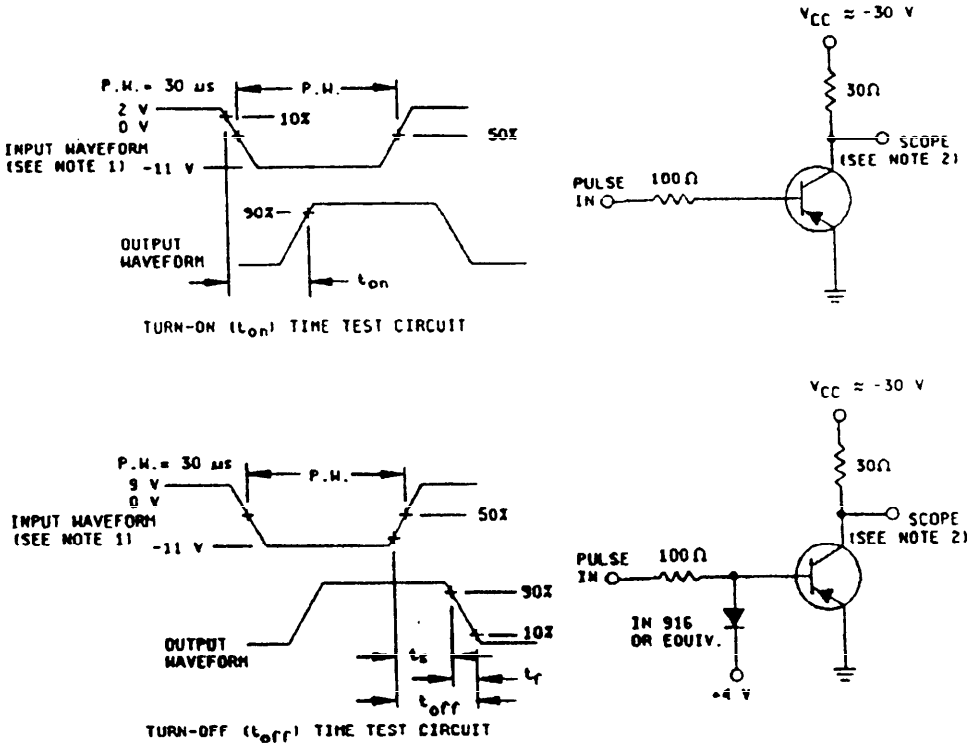
1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:
- a. Subgroup 3, table III, steps 1, 3 and 4.
- b. Subgroup 4, table III, steps 1, 3, 4, 7 and 8.
- c. Subgroup 5, table III, steps 1, 3, 4, 5, 6, 7 and 8.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:
- a. Subgroup 2, table III, steps 1 and 3.
- b. Subgroup 3, table III, steps 2, 3, and 8.
- c. Subgroup 6, table III, steps 2 and 3.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:
- a. Subgroup 2, table III, steps 1, 3, and 4 (JANS); steps 1 and 3 (JANTX and JANTXV).
- b. Subgroup 3, table III, steps 1, 3, and 4 (JANS); steps 1 and 3 (JANTX and JANTXV).
- c. Subgroup 6, table III, steps 1, 3, 4, 5, 6 and 8 (JANS); steps 2, 3, 6, and 8 (JANTX and JANTXV).

4/ Safe operating area (SOA) testing may be performed in lieu of thermal response testing herein provided that appropriate conditions of temperature, time, current, and voltage to achieve die attach integrity are submitted to the qualifying activity.

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 20 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 20 ns.

FIGURE 3. Pulse response test circuits.

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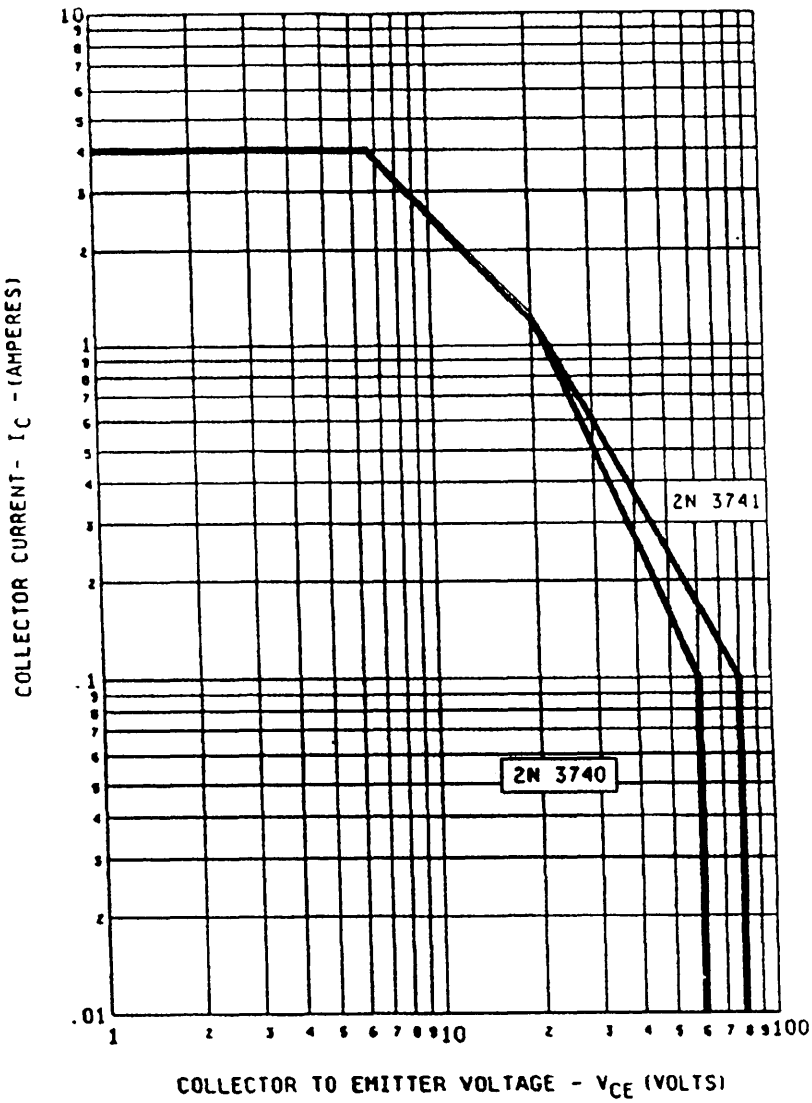
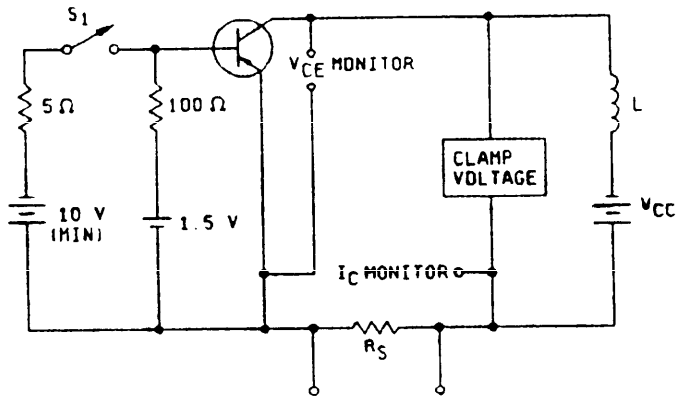


FIGURE 4. Maximum safe operating area graph (continuous dc).

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$R_S \leq .1\Omega$ 12 W; 1 percent tolerance max. (noninductive).

*L = 5 mH at 4 A with a maximum dc resistance of 0.5Ω. Reference only: 2 ESSEX Stancor C-2688 (in parallel), or equivalent.

Procedure

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if the clamp voltage is not reached.
3. Perform specified end-point tests.

FIGURE 5. Clamped inductive sweep test circuit.

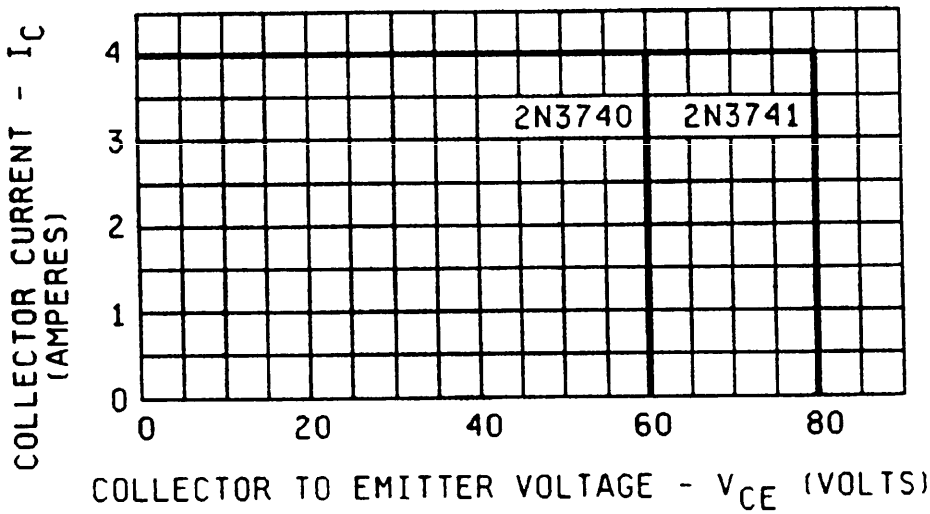


FIGURE 6. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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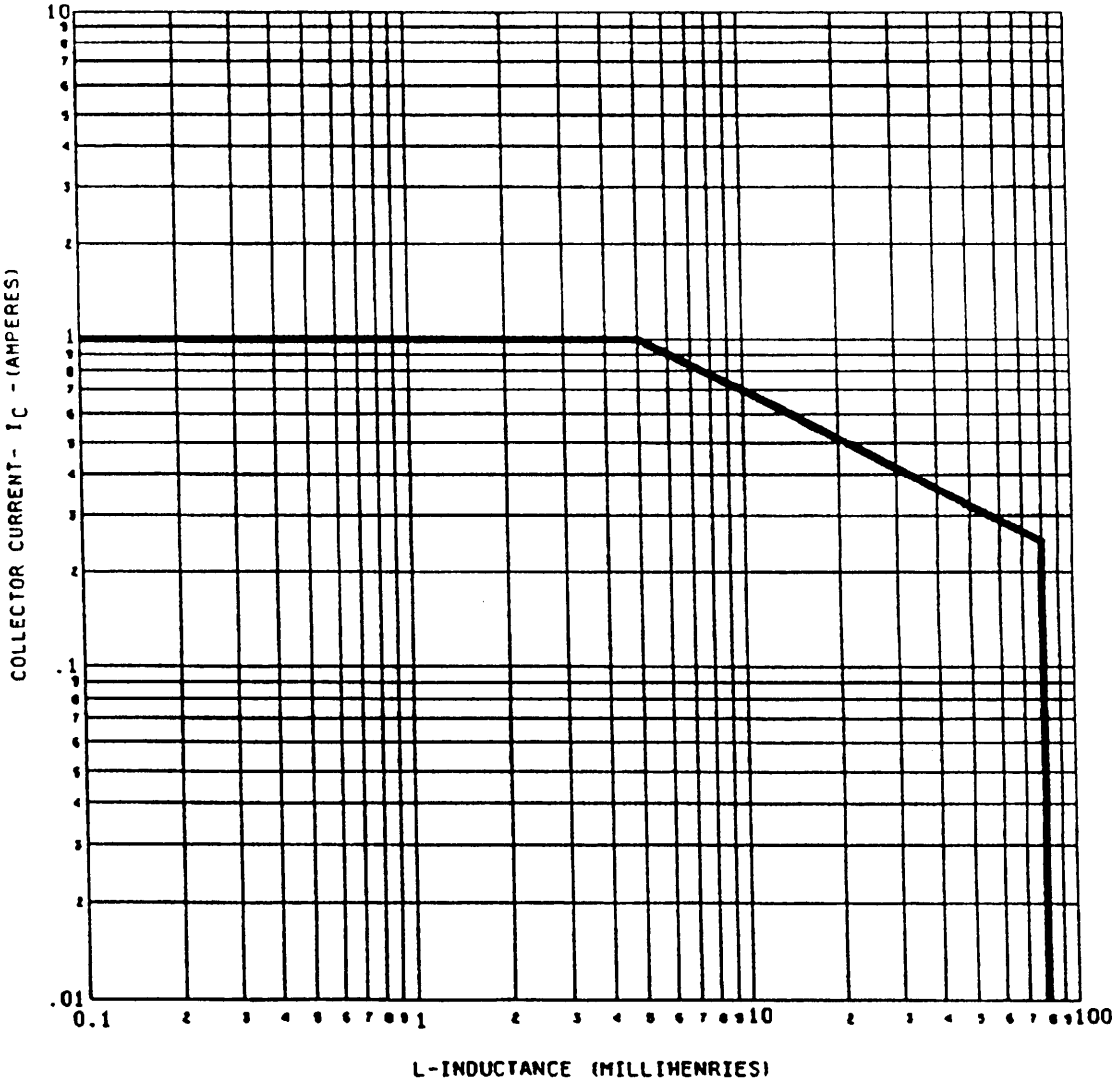


FIGURE 7. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition data. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead material and finish as specified (see 3.3.1).

c. For die acquisition, the JANC letter version shall be specified (see figure 2).

e. Type designation and product assurance level.

6.3 Replacement data. JANTX devices are a direct replacement for JAN devices (example, JANTX2N3740 for JAN2N3740).

6.4 Suppliers of JANC die. The qualified JANC suppliers with the applicable letter version (example JANCA3740) will be identified on the Qualified Products List.

JANC ordering information		
PIN	Manufacturer	
	33178	43611
2N3740	A3740	B3740
2N3741	A3741	B3741

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - AV, SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 15

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1441)

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/441D	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE Semiconductor Device, Transistor, PNP, Silicon, Power Types 2N3740 and 2N3741, JANTX, JANTXV, JANS, and JANC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	c. MAILING ADDRESS (Include Zip Code)

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c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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